# Graduate Application Dashboard System (GR.A.D.S.)

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## ABSTRACT

GRADS or the Graduate Application Dashboard system is an app that helps users keep track of their graduate school applications. This inspiration for this app came from my own personal needs as someone who is seeking to get their masters in Germany. GRADS is a platform specifically designed for keeping your applications organized and provides easy access to your application information and documents. The appeal of GRADS is that it offers more customization and features than using excel or google sheets. The emphasis of this app is to showcase my knowledge of C.R.U.D, MySQL, DBMS, and Java backend. Some new challenges that I will be taking on for this project are learning client-server architecture and web development, which I will learn as I am building this project. As well as learning the functionality of HTTP Request Methods such as GET, POST, PUT, and DELETE with REST APIs. A user will be able to sign into the application portal, create a new dashboard, and title it to their liking. For example, “Zack’s Germany Masters Application Dashboard”. Within this dashboard a user can add a new grad application to the dashboard and fill out the necessary fields, such as university name, program type, deadline, course URL, fees, and tuition. The new application gets displayed on the dashboard as a row. Users will be able to edit, delete, or add onto the existing application. Additional features will be added on later such as filtering, critical date countdowns and reminders, as well a place to store important application documents such as transcripts and recommendation letters.

## CURRENT/EXISTING IMPLEMENTATIONS

When looking online, I immediately found many GitHub pages of people making their own grad school application trackers. However, these repositories consist of markdown and README files which were manually written up. Also those repositories were essentially for personal use only and used for that

person's own use case. They don't use the same approach I am going after which makes this into a full stack application that can be used by others. As a result, I was not able to find anything close to what I am building.

## METHODOLOGY

I plan to accomplish this project by breaking down the development into incremental phases. This first phase consisted of designing and building the database, installing the Java framework Spring Boot along with the project's dependencies, connecting the backend to the MariaDB database, and finally initializing and pushing to my GitLab Repo. I started off first by creating an ERD diagram and a Relational Schema for the database. From there I used Dbeaver, a free open source DBMS tool to create the tables and columns with a SQL script. Next, I went onto start.spring.io to initialize a Spring Boot project. The project type I chose was maven, a project management tool. I also added the necessary dependencies that I needed which were Spring Data JPA, Spring Web, MySQL Driver, Thymeleaf, devtools, and lombok. From there, I downloaded the Spring Boot skeleton and opened it in IntelliJ. Then inside of IntelliJ I connected the Spring Boot with my local MariaDB. And finally, I initialized the Gitlab repository and then pushed it.

For this Project, I have chosen to use Java as my primary programming language. This is because Java is my strongest language. However, I have never worked with Spring Boot before or made a full on server-client based application. I have worked with HTML, CSS, and Javascript back in high school but I am rusty and need to refresh my skills. Because of this, I will primarily rely on free online resources, such as watching youtube tutorials, web documentation, and possibly using stackoverflow to troubleshoot any issues I may come across. This goal is to rely the least amount on AI as much as possible. As I will discuss in the disclosure section, I have used Claude to help me brainstorm and picture how I want this project to look like. The next step in my project is to complete the backend entities. So far, I have implemented the universities and application entities. I will need to implement the requirements and document entities next, completing the backend. Then to fully complete the backend, I will need to finish

adding all the proper endpoints to the entities controllers in my Spring Boot project. I will test and validate these endpoints using Postman. Once the backend has been fully implemented, then I will begin to work on the frontend.

## WHAT PROBLEM DOES YOUR TOOL/PRODUCT SOLVE IN REAL LIFE

The main problem that this project will solve is the issue of keeping applications organized. In this case, I will be building the project to accommodate Graduate schools applications since I actually plan to use this for my own purposes. However, this system could easily be modified to job and undergraduate applications. Of course many people tend to use excel or google sheets for this purpose but I want to provide more functionality and a clean, modern user friendly interface for users. What my application does is it allows users to modularize and store their applications and information in a relational style. Meaning that within each row of the dashboard, all relevant fields are listed and easily readable. Users can execute basic Create, Read, Update, and Delete operations (C.R.U.D.). I also plan on implementing unique features in my app that allows users to filter their dashboard based on the application status, have stats metrics for due dates, countdowns, and user specific data such as total number of applications completed. The main appeal I am working towards with this app is to make it data feedback focused so that users can see which application needs more priority or attention. For example, if an application still has not been submitted yet, there will be some red flag style marking to get the user's attention and call them to action to take care of it or delete it from their dashboard if they don't want to apply to that university anymore. The overall goal of this app is to utilize CRUD functions. However, just CRUD functions alone are not enough and too plain. This is where my unique features come in and add more value to the app and give greater reasons for users to use my app than just a storage list.

## FUTURE UPDATES/PLAN OF ACTION UNTIL THE SEMESTER ENDS

As I stated in the previous section, I am going to divide this project into phases. I am aware and expect that I will need to revisit previous phases in order to modify things like the database columns of the table if I find that I want to add, modify or delete something. As of where the project currently stands

in week 6, I plan on making steady progress everywhere following the 3 commits a week rule. I am very confident that this project will be completed well before the end of the semester and that I will have plenty of time to get more creative and add plenty of useful and attractive features. One more important thing to mention is that I am currently working on a log within this app called obsidian. The app uses markdown files, which I can use to create an organized and structured progress log of my project.

## PROJECT LINK

https://student-gitlab.sec.cnu.edu/zackary.siddiqi.22/grad-app-tracker

## MY DISCRETION SECTION

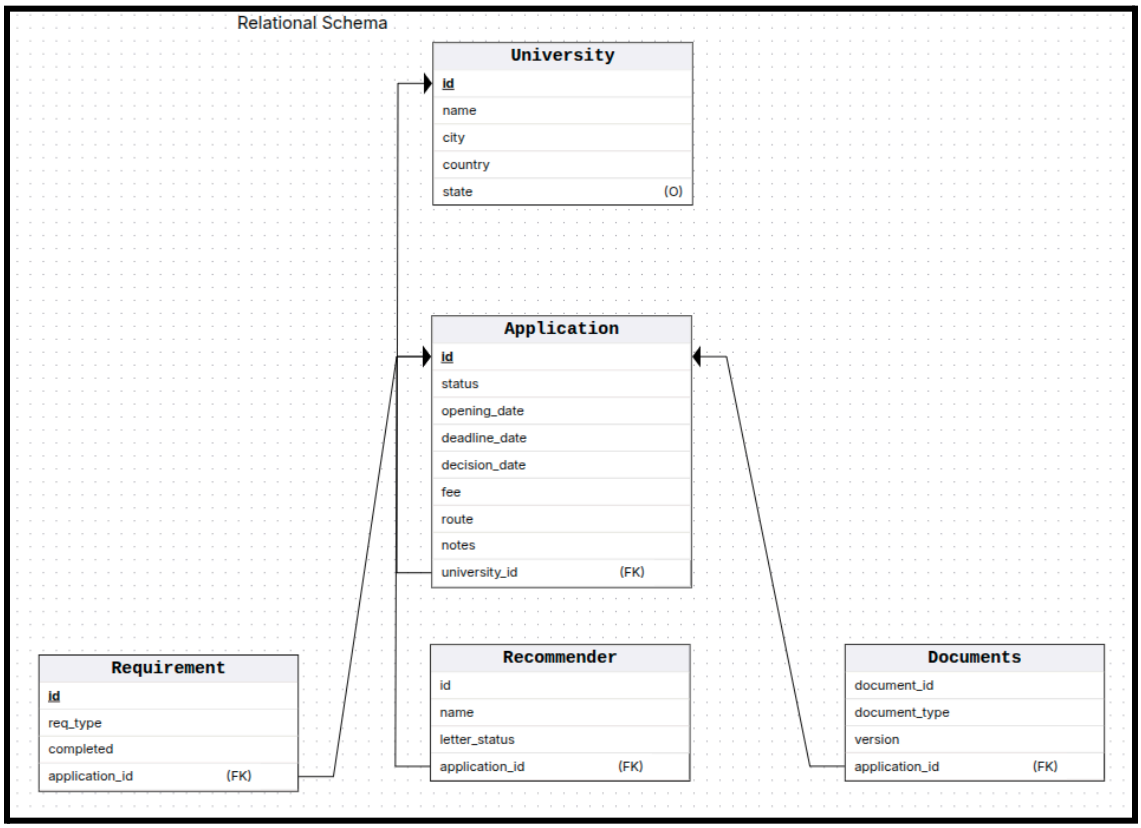
I want to be able to create a domain for the application and deploy it. There are many services that offer a free tier to do this. However, with my current tech stack there could be compatibility issues when I want to deploy my app. For example, I am currently not using react.js for my frontend and PostgreSQL for my DB. These tools are more commonly used when building full-stack applications. So, I will need to consider this during week 6 before I proceed further into my project.

## AI DISCLOSURE STATEMENT

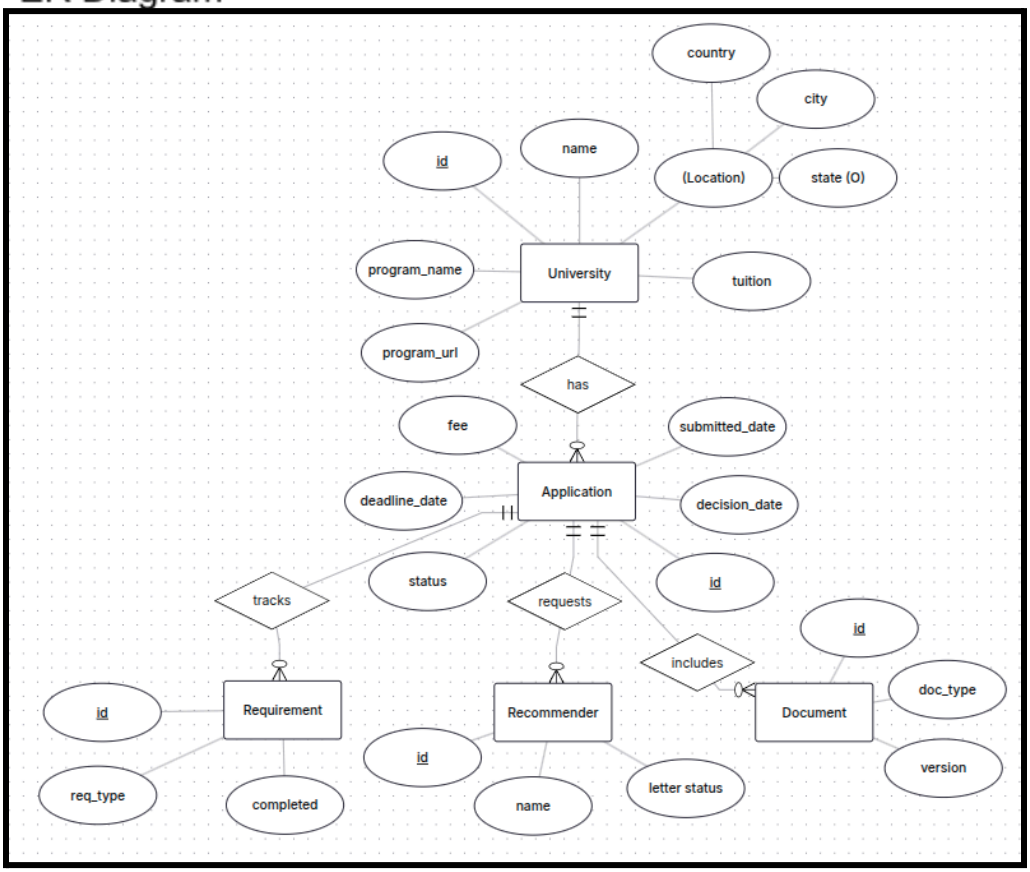
Starting at the beginning of the project, I used Claude in order to help start brainstorming a relevant project that would best suit my skill set. I told them that I wanted to create a project that involved using Java and MySQL, which is a very common combination for building full-stack apps. Claude then output many categories of projects that suited my case. However, it did not directly give me my idea. I noticed in the response to my prompt that Claude said creating an application that incorporated CRUD was a good option which sparked an idea for this Graduate application dashboard. Creating just a standard CRUD app is not enough and I know this, which is why the main highlight of the app will be the unique feature I add towards the end once I finish implementing the foundation. I also used Claude to help me figure out what kind of tools I will need. And it laid out the most applicable ones such as Spring Boot, Thymeleaf, MySQL, and HTML + CSS. For the code base, I have been following tutorials on Youtube to

help me learn the workings of Spring boot and the steps to initializing a project. I will use AI as a search engine to help explain certain concepts that I do not understand fully.  For example, what REST API endpoints are and HTTP requests. The main goal with AI for this project is to try and use it as little as possible since I want the project ideas and structural integrity to come from my own internal thinking process and my ideas. I understand now how easy it can be to fall into the trap of making AI do everything for you. And I want to prove in this project that I am fully capable of figuring out most of the project on my own and with the online resources. I believe the biggest challenge for this project is understanding how to implement the front end. I have very little experience with building frontend UIs, however I am taking this as an opportunity to integrate Full Stack development to my skillset. At this point in the project, AI will only be used as a search engine to help explain topics/concepts that I need help understanding in a generalized manner.

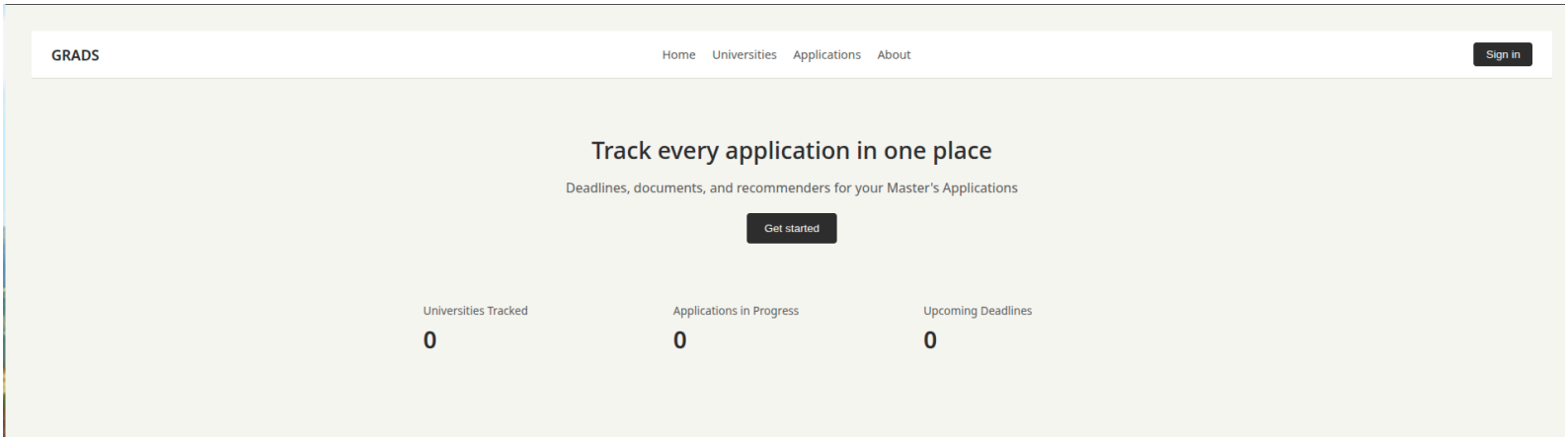
## PROJECT GRAPHICS



## ER Diagram




## Current Homepage



## REFERENCES


- https://youtu.be/olAva-QT0iw
- https://youtu.be/y3odhQtu4R8
- https://www.youtube.com/watch?v=99oZGQlMb0w&t=408s

*These tutorials aren't exactly what I'm doing with my project but layout the same foundation of what I'm trying to build*

- https://docs.spring.io/spring-boot/index.html
- https://start.spring.io/